Báo cáo tích hợp — Đặc tả → Kripke → Sinh mã → CWE/CVE

Đặc tả: smart_ev_teacher_e2e_v4

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1. Đặc tả yêu cầu (Specification)

Smart EV full E2E teacher verification v4. Ch1: V2 Kripke model (28 states, 47 transitions, 16 CTL properties). Ch2: 6 C functions — 2 safe baselines + 4 bug-injected probes (buffer-overflow, memory-leak, double-free, use-after-free). Ch3: 12 Z3 arithmetic checks covering SOC/TEMP/POWER/CHARGING domains with SAT/UNSAT pairs and overflow probes using production constants. Ch4: 3 Python functions — 2 divide (CWE-369) + 1 magic_value (CWE-798 concolic). Formal and SMT artifacts are grounded in Smart EV requirements and production constants. Ch2/Ch4 generated artifacts are capability probes linked to Smart EV scenarios and are not production-equivalent implementations.

Mã yêu cầu	Loại	Ưu tiên	Nội dung
REQ-AUTH-01	safety	HIGH	Vehicle authentication may succeed only when the vehicle is connected and the token is valid.
REQ-AUTH-02	safety	HIGH	When a vehicle disconnects from the station, authentication must be cleared.
REQ-SOC-01	safety	HIGH	Vehicle SOC must remain between 0% and 100%.
REQ-SOC-02	safety	HIGH	A vehicle with SOC at 100% must not continue receiving charging power.
REQ-TEMP-01	safety	HIGH	When battery temperature reaches or exceeds 60 C, charging must stop.
REQ-PORT-01	safety	HIGH	A port may enter CHARGING only when a valid vehicle is connected.
REQ-PORT-02	safety	HIGH	At any time, one charging port may be linked to at most one vehicle.
REQ-PORT-03	safety	HIGH	A faulted port must not deliver charging power.
REQ-CHG-01	safety	HIGH	Charging may start only when the vehicle is connected, authenticated, battery and temperature are safe, the port is ready, and no emergency or fault is active.

Mã yêu cầu	Loại	Ưu tiên	Nội dung
REQ-CHG-02	safety	HIGH	Charging requires the complete safety gate to remain valid.
REQ-EMG-01	safety	HIGH	Emergency or fault conditions must stop charging and force charging power to zero.

2. Kiểm chứng hình thức (Kripke/CTL)

Trạng thái deadlock: không có

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_auth_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_auth_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_soc_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_soc_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_temp_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation
AG(!req_port_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_port_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_port_03_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_chg_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_02_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
AG(!req_chg_02_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_emg_01_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation

Công thức CTL	Kết quả	Trạng thái thỏa
AG(!req_emg_01_violation)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL, unreachable_req_auth_01_violation, unreachable_req_auth_02_violation, unreachable_req_chg_01_violation, unreachable_req_chg_02_violation, unreachable_req_port_01_violation, unreachable_req_port_02_violation, unreachable_req_port_03_violation, unreachable_req_soc_01_violation, unreachable_req_soc_02_violation, unreachable_req_temp_01_violation
EF(charging)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT
EF(start_blocked)	ĐÚNG	AUTH_FAILED, CONNECTED, DISCONNECTED, START_BLOCKED
EF(stopped)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, DISCONNECTING, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, SOC_FULL, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL
EF(recovered)	ĐÚNG	AUTHENTICATED, CHARGING, CONNECTED, DISCONNECTED, EMERGENCY, OVERHEAT, PORT_FAULT, RECOVERED_SAFE, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT
AF(safe)	ĐÚNG	AUTHENTICATED, AUTH_FAILED, CHARGING, CONNECTED, DISCONNECTED, RECOVERED_SAFE, START_BLOCKED, STOPPED_DISCONNECT, STOPPED_EMERGENCY, STOPPED_OVERHEAT, STOPPED_PORT_FAULT, STOPPED_SOC_FULL

2.1. Sơ đồ chuyển trạng thái

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 1/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 1/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=blue!25] (nd_DISCONNECTED) at (0.36,2.15) {DISCONNECTED\\{\tiny
auth\_clear,disconnected,power\_zero,safe}};
\node[fill=gray!15] (nd_CONNECTED) at (-0.22,0.37) {CONNECTED\\{\tiny
connected,not\_authenticated,port\_occupied,power\_zero,safe}};
\node[fill=gray!15] (nd_AUTH_FAILED) at (-0.38,-0.77) {AUTH\_FAILED\\{\tiny
auth\_failed,connected,power\_zero,safe,start\_not\_allowed}};
\node[fill=gray!15] (nd_AUTHENTICATED) at (-2.20,0.66) {AUTHENTICATED\\{\tiny
authenticated,connected,port\_ready,power\_zero,safe}};
\node[fill=gray!15] (nd_START_BLOCKED) at (1.61,1.07) {START\_BLOCKED\\{\tiny
connected,power\_zero,safe,start\_blocked}};
\node[fill=gray!15] (nd_CHARGING) at (-0.70,-0.19) {CHARGING\\{\tiny
authenticated,charging,connected,power\_positive,safe,safety\_gate\_valid}};
\node[fill=gray!15] (nd_RECOVERED_SAFE) at (1.20,-1.00) {RECOVERED\_SAFE\\{\tiny
authenticated,connected,power\_zero,recovered,safe}};
\node[fill=gray!15] (nd_OVERHEAT) at (-0.92,-1.78) {OVERHEAT\\{\tiny
authenticated,connected,overheat,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_PORT_FAULT) at (-0.21,1.86) {PORT\_FAULT\\{\tiny
authenticated,connected,fault,port\_fault,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_DISCONNECTING) at (0.55,-1.58) {DISCONNECTING\\{\tiny
auth\_clear\_required,disconnecting,stop\_required}};
\node[fill=gray!15] (nd_SOC_FULL) at (0.32,0.09) {SOC\_FULL\\{\tiny
authenticated,connected,soc\_full,stop\_required,unsafe\_until\_stopped}};
\node[fill=gray!15] (nd_EMERGENCY) at (-1.38,-1.27) {EMERGENCY\\{\tiny
emergency,stop\_required,unsafe\_until\_stopped}};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_PORT_FAULT) at (1.23,1.00) {$\to$
STOPPED\_PORT\_FAULT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_OVERHEAT) at (-0.66,-0.67) {$\to$ STOPPED\_OVERHEAT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_EMERGENCY) at (0.86,-1.84) {$\to$
STOPPED\_EMERGENCY};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_DISCONNECT) at (1.79,-0.09) {$\to$
STOPPED\_DISCONNECT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__STOPPED_SOC_FULL) at (-1.25,1.99) {$\to$ STOPPED\_SOC\_FULL};
\draw (nd_DISCONNECTED) -- (nd_DISCONNECTED);
\draw (nd_DISCONNECTED) -- (nd_CONNECTED);
\draw (nd_CONNECTED) -- (nd_AUTH_FAILED);
\draw (nd_CONNECTED) -- (nd_CONNECTED);
\draw (nd_CONNECTED) -- (nd_AUTHENTICATED);
\draw (nd_AUTH_FAILED) -- (nd_START_BLOCKED);
\draw (nd_AUTH_FAILED) -- (nd_AUTH_FAILED);
\draw (nd_AUTHENTICATED) -- (nd_AUTHENTICATED);
\draw (nd_AUTHENTICATED) -- (nd_CHARGING);
\draw (nd_START_BLOCKED) -- (nd_START_BLOCKED);
\draw (nd_CHARGING) -- (nd_OVERHEAT);
\draw (nd_CHARGING) -- (nd_PORT_FAULT);
\draw (nd_CHARGING) -- (nd_DISCONNECTING);
\draw (nd_CHARGING) -- (nd_SOC_FULL);
\draw (nd_CHARGING) -- (nd_EMERGENCY);
\draw (nd_CHARGING) -- (nd_CHARGING);
\draw (nd_RECOVERED_SAFE) -- (nd_RECOVERED_SAFE);
\draw (nd_RECOVERED_SAFE) -- (nd_CHARGING);
\draw (nd_OVERHEAT) -- (nd_OVERHEAT);
\draw (nd_OVERHEAT) -- (nd___ghost__STOPPED_OVERHEAT);
\draw (nd_PORT_FAULT) -- (nd___ghost__STOPPED_PORT_FAULT);
\draw (nd_PORT_FAULT) -- (nd_PORT_FAULT);
\draw (nd_DISCONNECTING) -- (nd___ghost__STOPPED_DISCONNECT);
```

```
\draw (nd_DISCONNECTING) -- (nd_DISCONNECTING);
\draw (nd_SOC_FULL) -- (nd___ghost__STOPPED_SOC_FULL);
\draw (nd_SOC_FULL) -- (nd_SOC_FULL);
\draw (nd_EMERGENCY) -- (nd_EMERGENCY);
\draw (nd_EMERGENCY) -- (nd___ghost__STOPPED_EMERGENCY);
\draw (nd___ghost__STOPPED_PORT_FAULT) -- (nd_RECOVERED_SAFE);
\draw (nd___ghost__STOPPED_OVERHEAT) -- (nd_RECOVERED_SAFE);
\draw (nd___ghost__STOPPED_EMERGENCY) -- (nd_RECOVERED_SAFE);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 2/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 2/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_STOPPED_EMERGENCY) at (0.13,1.98) {STOPPED\_EMERGENCY\\{\tiny
emergency,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_PORT_FAULT) at (1.07,0.71) {STOPPED\_PORT\_FAULT\\{\tiny
authenticated,connected,fault,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_OVERHEAT) at (-1.49,-1.16) {STOPPED\_OVERHEAT\\{\tiny
authenticated,connected,overheat,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_DISCONNECT) at (-2.13,0.52) {STOPPED\_DISCONNECT\\{\tiny
auth\_clear,disconnected,power\_zero,safe,stopped}};
\node[fill=gray!15] (nd_STOPPED_SOC_FULL) at (1.76,1.36) {STOPPED\_SOC\_FULL\\{\tiny
authenticated,connected,power\_zero,safe,soc\_full,stopped}};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__RECOVERED_SAFE) at (-1.21,1.74) {$\to$ RECOVERED\_SAFE};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__EMERGENCY) at (2.13,-0.02) {$\to$ EMERGENCY};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__PORT_FAULT) at (0.46,-0.89) {$\to$ PORT\_FAULT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__OVERHEAT) at (-2.01,-0.85) {$\to$ OVERHEAT};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__DISCONNECTING) at (-0.91,-2.07) {$\to$ DISCONNECTING};
\node[rectangle, dashed, fill=gray!10] (nd___ghost__SOC_FULL) at (2.20,-1.32) {$\to$ SOC\_FULL};
\draw (nd_STOPPED_EMERGENCY) -- (nd_STOPPED_EMERGENCY);
\draw (nd_STOPPED_EMERGENCY) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_PORT_FAULT) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_PORT_FAULT) -- (nd_STOPPED_PORT_FAULT);
\draw (nd_STOPPED_OVERHEAT) -- (nd_STOPPED_OVERHEAT);
\draw (nd_STOPPED_OVERHEAT) -- (nd___ghost__RECOVERED_SAFE);
\draw (nd_STOPPED_DISCONNECT) -- (nd_STOPPED_DISCONNECT);
\draw (nd_STOPPED_SOC_FULL) -- (nd_STOPPED_SOC_FULL);
\draw (nd___ghost__EMERGENCY) -- (nd_STOPPED_EMERGENCY);
\draw (nd___ghost__PORT_FAULT) -- (nd_STOPPED_PORT_FAULT);
\draw (nd___ghost__OVERHEAT) -- (nd_STOPPED_OVERHEAT);
\draw (nd___ghost__DISCONNECTING) -- (nd_STOPPED_DISCONNECT);
\draw (nd___ghost__SOC_FULL) -- (nd_STOPPED_SOC_FULL);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 3/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 3/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_chg_02_violation) at (0.00,0.00) {unreachable\_req\_chg\_02\_violation\\{\tiny
req\_chg\_02,req\_chg\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_chg_02_violation) -- (nd_unreachable_req_chg_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 4/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 4/13
```

```
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_03_violation) at (0.00,0.00) {unreachable\_req\_port\_03\_violation\\{\tiny
req\_port\_03,req\_port\_03\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_03_violation) -- (nd_unreachable_req_port_03_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 5/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 5/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_01_violation) at (0.00,0.00) {unreachable\_req\_port\_01\_violation\\{\tiny
req\_port\_01,req\_port\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_01_violation) -- (nd_unreachable_req_port_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 6/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 6/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_temp_01_violation) at (0.00,0.00) {unreachable\_req\_temp\_01\_violation\\{\tiny
req\_temp\_01,req\_temp\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_temp_01_violation) -- (nd_unreachable_req_temp_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 7/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 7/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_auth_01_violation) at (0.00,0.00) {unreachable\_req\_auth\_01\_violation\\{\tiny
req\_auth\_01,req\_auth\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_auth_01_violation) -- (nd_unreachable_req_auth_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 8/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 8/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_port_02_violation) at (0.00,0.00) {unreachable\_req\_port\_02\_violation\\{\tiny
req\_port\_02,req\_port\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_port_02_violation) -- (nd_unreachable_req_port_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 9/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 9/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_soc_02_violation) at (0.00,0.00) {unreachable\_req\_soc\_02\_violation\\{\tiny req\_soc\_02,req\_soc\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_soc_02_violation) -- (nd_unreachable_req_soc_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 10/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 10/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_soc_01_violation) at (0.00,0.00) {unreachable\_req\_soc\_01\_violation\\{\tiny req\_soc\_01,req\_soc\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_soc_01_violation) -- (nd_unreachable_req_soc_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 11/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 11/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_auth_02_violation) at (0.00,0.00) {unreachable\_req\_auth\_02\_violation\\{\tiny req\_auth\_02,req\_auth\_02\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_auth_02_violation) -- (nd_unreachable_req_auth_02_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 12/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 12/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
\node[fill=gray!15] (nd_unreachable_req_chg_01_violation) at (0.00,0.00) {unreachable\_req\_chg\_01\_violation\\{\tiny req\_chg\_01,req\_chg\_01\_violation,smart\_ev\_v2,unreachable\_violation\_witness}};
\draw (nd_unreachable_req_chg_01_violation) -- (nd_unreachable_req_chg_01_violation);
\end{tikzpicture}
```

Sơ đồ: Mô hình chính: smart_ev_teacher_e2e_v4 — phần 13/13

Mã TikZ (LaTeX) — dán trực tiếp vào tài liệu LaTeX có \usepackage{tikz} để biên dịch thành hình vector, thay cho ảnh tĩnh:

```
% Mô hình chính: smart_ev_teacher_e2e_v4 — phần 13/13
\begin{tikzpicture}[
every node/.style={draw, circle, inner sep=2pt,
minimum size=8mm, align=center, font=\scriptsize},
->, >=stealth, thick, auto
]
```

\node[fill=gray!15] (nd_unreachable_req_emg_01_violation) at (0.00,0.00) {unreachable_req_emg_01_violation\\{\tiny req_emg_01,req_emg_01_violation,smart_ev_v2,unreachable_violation_witness}};
\draw (nd_unreachable_req_emg_01_violation) -- (nd_unreachable_req_emg_01_violation);
\end{tikzpicture}

3. Mã C sinh ra & Static Checker (lỗi bộ nhớ)

Mã C sinh từ đặc tả

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_copy_vehicle_id', kind='buffer_copy' */
void ev_copy_vehicle_id(const char *src_id) {
char buffer[20];
strncpy(buffer, src_id, sizeof(buffer) - 1);
buffer[sizeof(buffer) - 1] = '\0';
printf("%s\n", buffer);
}

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_copy_auth_token', kind='buffer_copy' */
void ev_copy_auth_token(const char *raw_token) {
char buffer[32];
strncpy(buffer, raw_token, sizeof(buffer) - 1);
buffer[sizeof(buffer) - 1] = '\0';
printf("%s\n", buffer);
}

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_parse_remote_cmd', kind='buffer_copy' */
void ev_parse_remote_cmd(const char *cmd_payload) {
char buffer[16];
strcpy(buffer, cmd_payload); /* sinh từ đặc tả: buffer-overflow */
printf("%s\n", buffer);
}

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_alloc_session_record', kind='alloc_release' */
void ev_alloc_session_record(void) {
int *buf = malloc(sizeof(int) * 64);
buf[0] = 0;
/* sinh từ đặc tả: memory-leak (thiếu free) */
}

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_release_port_context', kind='alloc_release' */
void ev_release_port_context(void) {
int *buf = malloc(sizeof(int) * 32);
buf[0] = 0;
free(buf);
free(buf); /* sinh từ đặc tả: double-free */
}

/* sinh từ đặc tả 'smart_ev_teacher_e2e_v4', hàm 'ev_resume_charging_after_fault', kind='alloc_release' */
void ev_resume_charging_after_fault(void) {
int *buf = malloc(sizeof(int) * 48);
buf[0] = 0;
free(buf);
buf[0] = 1; /* sinh từ đặc tả: use-after-free */
}
```

Dòng	Mức độ	Loại lỗi	Mô tả
24	HIGH	dangerous-function	Sử dụng hàm không an toàn 'strcpy()' — dễ gây buffer overflow. Goi y: dung strncpy thay the.

Dòng	Mức độ	Loại lỗi	Mô tả
30	MEDIUM	memory-leak	Con trỏ 'buf' được cấp phát (malloc/calloc/realloc) nhưng không thấy free() trong hàm ev_alloc_session_record() — nghi ngờ rò rỉ bộ nhớ
40	HIGH	double-free	Double-free trên con trỏ 'buf' (đã free() tại dòng 39) trong hàm ev_release_port_context()
48	HIGH	use-after-free	Khả năng sử dụng con trỏ 'buf' sau khi free() (freed tại dòng 47) trong hàm ev_resume_charging_after_fault()

4. Trích xuất AST/điều kiện & SMT (Z3)

Hàm	Biến	Điều kiện	Vòng lặp	Độ phức tạp McCabe	Độ sâu lồng nhau
ev_copy_vehicle_id	['buffer']	[]	[]	1	1
ev_copy_auth_token	['buffer']	[]	[]	1	1
ev_parse_remote_cmd	['buffer']	[]	[]	1	1
ev_alloc_session_record	['buf']	[]	[]	1	1
ev_release_port_context	['buf']	[]	[]	1	1
ev_resume_charging_after_fault	['buf']	[]	[]	1	1
ev_charging_power_ratio	[]	[]	[]	1	1
ev_token_validation_ratio	[]	[]	[]	1	1
ev_firmware_unlock_gate	[]	['unlock_code_a == 4721', 'unlock_code_b == 9033']	[]	3	3

Kiểm tra	Kết quả SMT	Model (nếu SAT)
gd12_soc_valid_domain_sat	sat	{'soc': 4}
gd12_soc_violation_contradiction_unsat	unsat	-
gd12_soc_update_no_overflow	sat	{'soc_current': 0, 'soc_delta': 0}
gd12_soc_update_no_overflow (int32 overflow?)	sat	{'soc_current': -1, 'soc_delta': -2147483648}
gd12_soc_full_threshold_sat	sat	{'soc': 100}
gd12_soc_charging_blocked_when_full	sat	{'soc': 100, 'charging': 0}
gd12_temp_overheat_threshold_sat	sat	{'temperature': 60}
gd12_temp_overheat_contradiction_unsat	unsat	-
gd12_temp_safe_range_sat	sat	{'temperature': 0, 'offset': 0}

Kiểm tra	Kết quả SMT	Model (nếu SAT)
gd12_temp_charging_blocked_when_overheat	sat	{'temperature': 60, 'charging': 0}
gd12_power_cap_not_exceeded	sat	{'power_port1': 0, 'power_port2': 0, 'power_port3': 0, 'power_port4': 0, 'power_port5': 0}
gd12_power_cap_not_exceeded (int32 overflow?)	sat	{'power_port1': -1, 'power_port2': 0, 'power_port3': -2147483648, 'power_port4': 0, 'power_port5': 0}
gd12_emergency_power_zero	sat	{'emergency': 1, 'total_power': 0}
gd12_charging_rate_positive_time	sat	{'energy_wh': 60060, 'time_seconds': 1}
gd12_charging_rate_positive_time (int32 overflow?)	sat	{'energy_wh': 1, 'time_seconds': 2147483647}

5. Dynamic Fuzzing (hàm Python sinh ra)

Hàm Python 'ev_charging_power_ratio'

```
def ev_charging_power_ratio(energy_delivered_wh, session_time_seconds):
return energy_delivered_wh // session_time_seconds
```

Hàm Python 'ev_token_validation_ratio'

```
def ev_token_validation_ratio(tokens_issued, tokens_active):
return tokens_issued // tokens_active
```

Hàm Python 'ev_firmware_unlock_gate'

```
def ev_firmware_unlock_gate(unlock_code_a, unlock_code_b):
if unlock_code_a == 4721:
if unlock_code_b == 9033:
raise ValueError('magic value reached')
return unlock_code_a + unlock_code_b
```

Hàm	Kết quả fuzzing
ev_charging_power_ratio	black-box: 11/300 lan crash
ev_token_validation_ratio	black-box: 11/300 lan crash
ev_firmware_unlock_gate	concolic: crash_input={'unlock_code_a': 4721, 'unlock_code_b': 9033} sau 3 lan thu

6. Phân loại CWE/CVE tổng hợp

Nguồn	Vị trí	Loại phát hiện	CWE	Ví dụ CVE minh họa
ch2	dong 24	dangerous-function	CWE-120 — Buffer Copy without Checking Size of Input	CVE-2003-0352 (2003)
ch2	dong 30	memory-leak	CWE-401 — Missing Release of Memory after Effective Lifetime	-
ch2	dong 40	double-free	CWE-415 — Double Free	-
ch2	dong 48	use-after-free	CWE-416 — Use After Free	CVE-2019-0708 (2019)
ch3	gd12_soc_update_no_overflow	integer-overflow	CWE-190 — Integer Overflow or Wraparound	CVE-2016-5195 (2016)
ch3	gd12_power_cap_not_exceeded	integer-overflow	CWE-190 — Integer Overflow or Wraparound	CVE-2016-5195 (2016)
ch3	gd12_charging_rate_positive_time	integer-overflow	CWE-190 — Integer Overflow or Wraparound	CVE-2016-5195 (2016)
ch4	ev_charging_power_ratio	uncontrolled-crash	CWE-248 — Uncaught Exception	-
ch4	ev_token_validation_ratio	uncontrolled-crash	CWE-248 — Uncaught Exception	-
ch4	ev_firmware_unlock_gate	uncontrolled-crash	CWE-248 — Uncaught Exception	-

7. Test case truy vết từ đặc tả (tài liệu hóa)

ID	Ưu tiên	Tiêu đề	Truy vết yêu cầu	Kết quả mong đợi
GD12-TC-CH1-CTL-16	HIGH	Ch1 verifies preserved Smart EV V2 Kripke model — 16 CTL properties	REQ-AUTH-01, REQ-AUTH-02, REQ-SOC-01, REQ-SOC-02, REQ-TEMP-01, REQ-PORT-01, REQ-PORT-02, REQ-PORT-03, REQ-CHG-01, REQ-CHG-02, REQ-EMG-01	No deadlock. All 16 CTL properties DUNG.
GD12-TC-CH2-SAFE-CODEGEN	MEDIUM	Ch2 static checker: safe buffer-copy functions produce zero findings	-	Zero Ch2 findings. TOOL_CAPABILITY_EVIDENCE / PROBE_ONLY confirmed; not production REQ PASS evidence.
GD12-TC-CH2-UNSAFE-CODEGEN	HIGH	Ch2 static checker: unsafe functions trigger CWE-121/401/415/416	-	Four CWE findings: CWE-121, CWE-401, CWE-415, CWE-416. Classification: BUG_INJECTED_PROBE / PROBE_ONLY; not production REQ PASS/FAIL evidence.
GD12-TC-CH3-SOC-CONTRACT	HIGH	Ch3 Z3: SOC domain contract — 5 SMT checks	REQ-SOC-01, REQ-SOC-02	SAT: gd12_soc_valid_domain_sat, gd12_soc_update_no_overflow, gd12_soc_full_threshold_sat, gd12_soc_charging_blocked_when_full. UNSAT: gd12_soc_violation_contradiction_unsat. No CWE-190 from gd12_soc_update_no_overflow (bounded by [0,100]).
GD12-TC-CH3-TEMP-CONTRACT	HIGH	Ch3 Z3: Temperature domain contract — 4 SMT checks	REQ-TEMP-01	SAT: overheat_threshold, safe_range, charging_blocked. UNSAT: contradiction. All 4 checks pass.
GD12-TC-CH3-POWER-CONTRACT	HIGH	Ch3 Z3: Power and station capacity contract — 3 SMT checks	REQ-CHG-01, REQ-EMG-01	All 3 SAT. No int32 overflow within production bounds. Power cap and emergency semantics confirmed satisfiable.

ID	Ưu tiên	Tiêu đề	Truy vết yêu cầu	Kết quả mong đợi
GD12-TC-CH4-FUZZ-DIVIDE	HIGH	Ch4 black-box fuzz: division-by-zero on charging rate and token ratio	-	Both functions crash with ZeroDivisionError when denominator=0. CWE-369 classified as PROBE_ONLY. Demonstrates Ch4 pipeline runs end-to-end; not production REQ evidence.
GD12-TC-CH4-CONCOLIC-MAGIC	MEDIUM	Ch4 concolic fuzz: firmware unlock gate with Z3-guided input synthesis	-	Concolic fuzzer finds crash_input={unlock_code_a:4721, unlock_code_b:9033}. CWE-798 finding generated as PROBE_ONLY, not a confirmed production defect.
GD12-TC-E2E-TRACEABILITY	HIGH	E2E traceability: requirement -> production evidence, with Ch2/Ch4 separated as tool capability	REQ-AUTH-01, REQ-SOC-01, REQ-SOC-02, REQ-TEMP-01, REQ-CHG-01, REQ-EMG-01	Full traceability chain: each requirement has at least one chapter finding. CWE findings traceable to requirement IDs.